

boot_android_emulator

boot_android_emulator.sh

Field	Value
Revision	1
Last modified	2026-06-21T17:50:00Z
Status	active
§11.4.18	In-source doc block present in scripts/ boot_android_emulator.sh

Overview

Boots an Android AVD on a remote Linux host (`nezha.local`) through the `basic-digital/containers` submodule (§11.4.76) — or with a clean direct fallback when the submodule’s Go tooling is not built for the target.

Prerequisites

- SSH key-based access to the remote host
- Android SDK + desired AVD installed on the remote host (see `EMULATED_DEVICE_TESTING.md`)
- `libbsd.so.0` on the remote host (emulator dependency)
- Containers submodule checked out at `containers/` (project root sibling)

Usage

```
# Default boot (CZ_API36_Phone on nezha.local)
```

```
bash scripts/boot_android_emulator.sh
```

```
# Custom AVD and host
```

```
SSH_HOST=other-host.local AVD=Pixel_9a bash scripts/  
boot_android_emulator.sh
```

```
# Custom port and resources
```

```
PORT=5556 RAM_MB=4096 CORES=4 bash scripts/  
boot_android_emulator.sh
```

Environment Variables

Variable	Default	Description
SSH_HOST	nezha.local	Remote host
SSH_USER	milosvasic	SSH user
AVD	CZ_API36_Phone	AVD name
PORT	5554	Emulator console port
ANDROID_SDK_ROOT	/home/milosvasic/ Android/Sdk	SDK path on remote
LD_LIBRARY_PATH_EXTRA	/home/ milosvasic/.local/ lib	Extra lib path (libbsd)
RAM_MB	3072	Emulator RAM
CORES	2	Emulator CPU cores
GPU_MODE	swiftshader_indirect	GPU acceleration
COLD_BOOT	true	Force cold boot (wipe)
BOOT_TIMEOUT_SEC	180	Boot timeout

Outputs

- **Evidence directory:** docs/qa/<run-id>-android-emu-boot/
- **Attestation:** attestation.json with device properties + boot metrics
- **State file:** emu_state.env for consumption by other scripts
- **SSH tunnel:** ADB available at localhost:<PORT>

Cleanup

The script does NOT auto-kill the emulator (to allow multiple sessions to use the running instance). To stop:

```
ssh nezha.local "ps aux | grep qemu | grep -v grep | awk '{print \
$2}' | xargs -r kill"
```

Or use the containers submodule's emulator-cleanup command when built.

Cross-references

- [docs/design/EMULATED_DEVICE_TESTING.md](#)
- [containers/pkg/emulator/](#)
- [containers/cmd/emulator-matrix/](#)
- [containers/images/android-test/](#)
- [scripts/boot_android_emulator.sh](#)
- §11.4.76 Containers-submodule mandate
- §11.4.161 Rootless container runtime mandate

Last verified

2026-06-21 — boot on nezha.local (Linux x86_64, KVM, Android API 36).